

Software Design Principles Exam Questions and Answers

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Question 1: Hospital System

Consider a hospital system.

1. Introduce three actors.
2. Design a Use Case with an **extend** relationship and explain why **extend** was chosen.
3. Describe an Activity Diagram process that has at least one Loop.

Answer:

1. **Introducing Actors:** Patient, Doctor, Receptionist.
2. **Designing a Use Case with an extend relationship:**
 - **Main Use Case:** Examine Patient
 - **Optional Use Case:** Request Lab, which is connected to Examination via an **extend** relationship.
 - **Reason for choosing extend:** Requesting a lab test is not an invariable part of the examination process and is performed only if the doctor diagnoses it (under specific conditions). Therefore, it is an optional process, and **extend** is suitable for it.
3. **Describing an Activity Diagram process (with a loop):**
 1. **Start**
 2. Patient visits reception.
 3. Patient information is registered by the receptionist.
 4. **Decision:** Is the patient's information complete?
 - If **No**: Return to step 3 (**This is the Loop**).
 - If **Yes**: Continue process.
 5. Assign an appointment and send the patient to the doctor.
 6. Patient examination by the doctor.
 7. **Decision:** Is a lab test required?

- If **Yes**: Register lab request $\leftarrow$ Receive lab result $\leftarrow$ Continue process.
 - If **No**: Continue process.
8. Register prescription and end visit.
 9. **End**

Question 2: Process Models

Name prescriptive process models and explain the incremental process model.

Answer:

Prescriptive process models include: Waterfall, Incremental, Evolutionary, and Concurrent development models.

Explanation of the Incremental Model: This model is used when initial requirements are reasonably well-defined, but constraints do not allow the entire project to be built uniformly all at once. In this model, several increments (development stages) are executed. Each increment encompasses a complete cycle of development stages, and at the end of each increment, a deliverable version (Increment) is provided. Each increment is used and evaluated by the customer, and feedback is applied in subsequent increments.

Question 3: Framework Activities

Name the framework activities for software development.

Answer:

Framework activities in software development comprise five main sections:

1. **Communication:** In this section, the development team communicates with the customer or stakeholders to accurately understand, clarify, and gather requirements and expectations to achieve a better understanding of needs.
2. **Planning:** The roadmap or execution path of the project is specified; including time and cost estimation, definition of technical tasks, and project scheduling.
3. **Modeling:** Gathered requirements are transformed into understandable models to gain a better grasp of software features, such as architectural design and UML diagrams.
4. **Construction:** Coding is performed in this phase, and tests are executed to identify and fix bugs to produce the software product.
5. **Deployment:** The final or intermediate product is delivered to customers, and their feedback is received.

Note: In many projects, these activities are performed iteratively as the project progresses.

Question 4: Gym Management System

Consider a gym management system with the following features:

- Individuals can register and become gym members.
- Members can reserve classes.
- Trainers can record member attendance.
- The system automatically creates a waiting list if a class reaches full capacity.
- Class fee payment can be online or in-person.
- The club manager generates a monthly income report.
- If a class is canceled, waiting list members are notified.

Identify the actors, extract use cases, and specify include/extend relationships.

Answer:

1. Identifying Actors:

- Visitor / Prospect
- Club Member
- Trainer
- Club Manager
- System - for automated tasks
- Payment Gateway

2. Extracting Use Cases:

- Sign up / Join
- Reserve Class
- Record Attendance
- Online Checkout
- Offline Checkout
- Cancel Reservation
- Join Waiting List
- Notify Waiting List Members
- Generate Monthly Income Report

3. Include and Extend Relationships:

- **Reserve Class** → *include* → **Check Class Capacity**: For a reservation, capacity must always be checked (mandatory).

- **Join Waiting List** → *extend* → **Reserve Class**: If capacity is full during reservation, joining the waiting list is optionally triggered.
- **Cancel Reservation** → *include* → **Check Cancel Time Rule**: Upon cancellation, time rules (cancellation time limit) must be checked.
- **Notify Waiting List** → *extend* → **Cancel Reservation**: Only when a reservation is canceled does the system notify the waiting list (under specific conditions).
- **Generate Monthly Report** → *include* → **Calculate Monthly Income**: For reporting, income and expense calculations must be performed.